

GearDesign Report

Cylindrical gear(pair)

작성일: 2025년 10월 17일

Contents

1. Summary

- 1.1. Gear geometry
- 1.2. Total safety factor
- 1.3. Total damage & life
- 1.4. Others
- 1.5. Summary - Load cases

2. Detail geometry

- 2.1. Input gear geometry
- 2.2. Calculated geometry - Gear
- 2.3. Calculated geometry - Tooth
- 2.4. Calculated geometry - Mesh
- 2.5. Calculated geometry - Accuracy
- 2.6. Calculated geometry - Measurement

3. Detail rating

- 3.1. General
- 3.2. Pitting
- 3.3. Bending
- 3.4. Scuffing
- 3.5. Micro-pitting

4. Power loss

- 4.1. Mesh loss
- 4.2. Churning loss
- 4.3. Windage loss

1. Summary

1.1. Gear geometry

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal module (mm)	m_n	2.5000	
Pressure angle at normal section (deg)	α_n	20.0000	
Helix angle at reference circle (deg)	β	0.0000	
Manual normal backlash (mm)	j_{bn}	0.0000	
Number of teeth	z	30.0000	41.0000
Facewidth (mm)	b	45.0000	45.0000
Profile shift coefficient	x	0.3132	0.2129
Quality (ISO)	Q	6.0000	6.0000
Center distance (mm)	a	90.0020	
Tooth thickness tolerance	-	DIN 3967 cd25	DIN 3967 cd25
Gear material	-	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)
Reference diameter (mm)	d	75.0000	102.5000
Tip diameter (mm)	d_a	81.566 (81.566 / 81.466)	108.564 (108.564 / 108.464)
Root diameter (mm)	d_f	70.316 (70.124 / 70.014)	97.314 (97.122 / 97.012)
Number of teeth spanned	k	4.0000	5.0000
Base tangent length (no backlash) (mm)	W_k	27.417 (27.351 / 27.314)	35.011 (34.945 / 34.908)
Effective Diameter of ball/pin (mm)	D_M	5.0000	4.5000
Diametral measurement over two balls (mm)	M_{dk}	84.529 (84.386 / 84.305)	110.031 (109.870 / 109.778)

1.2. Total safety factor

	Gear1 [Left]	Gear1 [Right]	Gear2 [Left]	Gear2 [Right]	Required S.F
Contact S.F	0.5958		0.6113		1.0000
Bending S.F	0.6163		0.6092		1.4000
Scuffing S.F (Flash Temp.)	3.1081				2.0000
Micropitting S.F	0.3099				1.0000

1.3. Total damage & life

	Gear1 [Left]	Gear1 [Right]	Gear2 [Left]	Gear2 [Right]
Contact Damage [%]	9999.9000	0.0000	9999.9000	0.0000
Bending Damage [%]	9999.9000	0.0000	9999.9000	0.0000
Contact Life [hr]	8.0000	∞	13.0000	∞
Bending Life [hr]	0.2000	∞	0.2000	∞
Total Req. Life [hr]	10000.0000			

1.4. Others

	Gear1 - Gear2
Transverse Contact Ratio	1.5337
Overlap Ratio	0.0000
Total Contact Ratio	1.5337
Worst Pitch Line Velocity [m/s]	3.9824
Total Efficiency [%]	99.1289
Worst Mesh Loss [W]	970.0643
Worst Churning Loss [W]	31.1088
Worst Peak-to-Peak T.E. [μm]	18.4501
Gear Total Mass [kg]	1.8737
Hunting tooth	Complete

1.5. Summary - Load cases

[illegible]

2. Detail geometry

2.1. Input gear geometry

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal module (mm)	m_n	2.5000	
Pressure angle at normal section (deg)	α_n	20.0000	
Helix angle at reference circle (deg)	β	0.0000	
Manual normal backlash (mm)	j_{bn}	0.0000	
Number of teeth	z	30.0000	41.0000
Facewidth (mm)	b	45.0000	45.0000
Profile shift coefficient	x	0.3132	0.2129
Quality (ISO)	Q	6.0000	6.0000
Center distance (mm)	a	90.0016	
Tooth thickness tolerance	-	DIN 3967 cd25	DIN 3967 cd25
Addendum coefficient	h^*_aP	1.0000	1.0000
Dedendum coefficient	h^*_fP	1.2500	1.2500
Root radius coefficient	ρ^*_fP	0.3800	0.3800
Tip form height coefficient	h^*_{FaP}	0.9600	0.9600
Ramp angle (deg)	α_{KP}	50.0000	50.0000
Protuberance height coefficient	h^*_{prP}	0.0000	0.0000
Protuberance angle (deg)	α_{prP}	0.0000	0.0000
Tip alteration (mm)	k_{m_n}	0.0000	0.0000
Tip radius coefficient (for topping)	ρ^*_aP	0.0000	0.0000
Gear material	-	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)	DIN 18CrNiMo7 ($\sigma_{Hlim} = 430$ Mpa)
Young's modulus (N/mm ²)	E	2.0600×10^5	2.0600×10^5
Poisson's ratio	ν	0.3000	0.3000
Density (kg/m ³)	ρ	7830.0000	7830.0000
Coeff. of thermal expansion (10 ⁻⁸ /°C)	α_{coeff}	11.5000	11.5000
Surface hardness	HRC	61.0000	61.0000
Core hardness	HBW	325.0000	325.0000
Tensile strength (N/mm ²)	σ_B	1200.0000	1200.0000
Yield strength (N/mm ²)	σ_s	850.0000	850.0000
Specific heat capacity (J/kg.K)	c_M	485.0000	485.0000
Specific heat conductivity (W/m.K)	λ_M	50.0000	50.0000
Mean roughness height, root (um)	R_{ZF}	20.0000	20.0000
Mean roughness height, flank (um)	R_{ZH}	4.8000	4.8000
Roughness average value, root (um)	R_{AF}	3.0000	3.0000
Roughness average value, flank (um)	R_{AH}	0.6000	0.6000

	Symbol	Pair1 Gear1	Pair1 Gear2
Fatigue strength for bending stress (N/mm ²)	σ_{Flim}	430.0000	430.0000
Fatigue strength for contact stress (N/mm ²)	σ_{Hlim}	1500.0000	1500.0000

2.2. Calculated geometry - Gear

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Gear ratio	GR	1.3667	
Transverse module (mm)	m_t	2.5000	
Working normal pressure angle (deg)	α_{wn}	22.0853	
Transverse pressure angle (deg)	α_t	20.0000	
Working transverse pressure angle (deg)	α_{wt}	22.0853	
Working transverse pressure angle, e/i (deg)	$\alpha_{wt}, e/i$	22.1026 / 22.0681	
Base helix angle (deg)	β_b	0.0000	
Helix angle at operating pitch circle (deg)	β_w	0.0000	
Radial backlash without tolerance (mm)	j_{r0}	0.0000	
Circumferential backlash without tolerance (mm)	j_{t0}	0.0000	
Min/max circumferential backlash by flank tolerance for each gear (mm)	$j_t, e/i$	0.0700 / 0.1100	0.0700 / 0.1100
Min/max normal backlash by flank tolerance for each gear (mm)	$j_n, e/i$	0.0658 / 0.1034	0.0658 / 0.1034
Circumferential backlash, e/i (mm)	$j_{tw}, e/i$	0.1330 / 0.2320	
Normal backlash, e/i (mm)	$j_{nw}, e/i$	0.1233 / 0.2150	
Angular backlash, min (deg)	j_θ, min	0.2005	0.1467
Angular backlash, max (deg)	j_θ, max	0.3496	0.2558
Effective facewidth (mm)	b_{eff}	45.0000	
Sum of the profile shift coefficients	Sum_x	0.5261	
Generating profile shift coefficient	$x_E, e/i$	0.2747 / 0.2528	0.1744 / 0.1524
Prevent undercut profile shift coefficient	x_{Emin}	-0.7547	-1.3981
Max. profile shift coeff. (min top land)	x_{Emax}	1.2935	1.6118
Reference center distance, $x_1=x_2=0$ (mm)	a_d	88.7500	
Working center distance (mm)	a_w	90.0016	
Reference diameter (mm)	d	75.0000	102.5000
Operating pitch diameter (mm)	d_w	76.0577	103.9455
Operating pitch diameter with tolerance (mm)	$d_w, e/i$	76.0670 / 76.0484	103.9582 / 103.9328
27		81.5660	108.5643
Tip diameter with tolerance (mm)	$d_a, e/i$	81.5660 / 81.4660	108.5643 / 108.4643

	Symbol	Pair1 Gear1	Pair1 Gear2
Root diamter (mm)	d_f	70.3160	97.3143
Root diameter with tolerance (mm)	d_f, e/i	70.1237 / 70.0138	97.1220 / 97.0121
Tip form diameter (mm)	d_Fa	81.3660	108.3643
Tip form diameter with tolerance (mm)	d_Fa, e/i	81.1457 / 81.0202	108.1516 / 108.0304
Root form diamter (mm)	d_Ff	72.1853	99.1558
Root form diameter with tolerance (mm)	d_Ff, e/i	72.0658 / 71.9994	99.0237 / 98.9496
Active tip diameter (EAP) (mm)	d_Na	81.3660	108.3643
Active tip diameter with tolerance (mm)	d_Na, e/i	81.1457 / 81.0202	108.1516 / 108.0304
Active root diameter (SAP) (mm)	d_Nf	72.7453	100.0357
Active root diameter with tolerance (mm)	d_Nf, e/i	72.9454 / 72.8473	100.2422 / 100.1401
Mean diameter for mass calculation (mm)	d_m	75.7924	102.7907
Gear mass (kg)	m	0.7661	1.1076
Gear inertia (kg.m^2)	J	0.0000	0.0000
Nominal tip clearance (mm)	c	0.5614	0.5614
Effective tip clearance, e/i (mm)	c, e/i	0.7735 / 0.6466	0.7735 / 0.6466
Min. margin of involute contact, (d_Nf_i - d_Ff_e) / 2 (mm)	c_Nff	0.3907	0.5582

2.3. Calculated geometry - Tooth

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Tooth depth (mm)	h	5.6250	5.6250
Tooth depth, e/i (mm)	h, e/i	5.7761 / 5.6712	5.7761 / 5.6712
Working depth (mm)	h_w	5.0636	
Addendum (mm)	h_a	3.2830	3.0322
Addendum, e/i (mm)	h_a, e/i	3.2830 / 3.2330	3.0322 / 2.9822
Dedendum (mm)	h_f	2.3420	2.5928
Dedendum, e/i (mm)	h_f, e/i	2.4382 / 2.4931	2.6890 / 2.7440
Number of virtual gear teeth	z_n	30.0000	41.0000
Tooth thickness coeff.	s_nc	1.7988	1.7257
Transverse tooth thickness (mm)	s_t	4.4970	4.3144
Normal tooth thickness (mm)	s_n	4.4970	4.3144
Normal tooth thickness at d_a (mm)	s_an	1.6139	1.7920
Normal tooth thickness at d_a, e/i (mm)	s_an, e/i	1.5940 / 1.4943	1.7682 / 1.6755
Normal tooth thickness at d_Fa (mm)	s_Fan	1.7258	1.8922
Normal tooth thickness at d_Fa, e/i (mm)	s_Fan, e/i	1.8402 / 1.7283	1.9832 / 1.8814
Normal space width at d_f (mm)	e_fn	0.0000	2.0061
Normal space width at d_f, e/i (mm)	e_fn, e/i	0.0000 / 0.0000	2.0422 / 2.0640

	Symbol	Pair1 Gear1	Pair1 Gear2
Normal pitch (mm)	p_n	7.8540	
Transverse pitch (mm)	p_t	7.8540	
Normal base pitch (mm)	p_bn	7.3803	
Transverse base pitch (mm)	p_bt	7.3803	
Axial pitch (mm)	p_x	0.0000	
Lead (mm)	p_z	0.0000	0.0000

2.4. Calculated geometry - Mesh

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Length of the path of contact (A to E) (mm)	g_α	11.3190	
Length of path of contact, e/i (mm)	g_α, e/i	10.8940 / 10.5750	
Length of approach POC of pinion (A to C) (mm)	g_f1	5.2862	
Length of approach POC of pinion, e/i (mm)	g_f1, e/i	5.0702 / 5.0364	
Length of recess POC of pinion (C to E) (mm)	g_a1	6.0328	
Length of recess POC of pinion, e/i (mm)	g_a1, e/i	5.8239 / 5.7992	
Length T1 - A (mm)	T1A	9.0122	
T1A tolerance (mm)	e/m/i	9.4079 / 9.3119 / 9.2158	
Length T1 - B (mm)	T1B	12.9508	
T1B tolerance (mm)	e/m/i	12.7295 / 12.6661 / 12.6026	
Length T1 - C (mm)	T1C	14.2983	
T1C tolerance (mm)	e/m/i	14.3107 / 14.2983 / 14.2860	
Length T1 - D (mm)	T1D	16.3925	
T1D tolerance (mm)	e/m/i	16.7882 / 16.6922 / 16.5961	
Length T1 - E (mm)	T1E	20.3312	
T1E tolerance (mm)	e/m/i	20.1099 / 20.0464 / 19.9829	
Length T2 - A (mm)	T2A	24.8272	
T2A tolerance (mm)	e/m/i	24.5943 / 24.5275 / 24.4608	
Length T2 - B (mm)	T2B	20.8886	
T2B tolerance (mm)	e/m/i	21.2661 / 21.1733 / 21.0806	
Length T2 - C (mm)	T2C	19.5411	
T2C tolerance (mm)	e/m/i	19.5580 / 19.5411 / 19.5242	
Length T2 - D (mm)	T2D	17.4469	
T2D tolerance (mm)	e/m/i	17.2140 / 17.1472 / 17.0804	
Length T2 - E (mm)	T2E	13.5082	
T2E tolerance (mm)	e/m/i	13.8857 / 13.7930 / 13.7003	
Length T1 - T2 (mm)	T1T2	33.8394	

	Symbol	Pair1 Gear1	Pair1 Gear2
T1T2 tolerance (mm)	e/m/i	33.8687 / 33.8394 / 33.8101	
Diameter of single contact point A (mm)	d_A	72.7453	108.3643
Diameter of single contact point B (mm)	d_B	75.0859	104.9885
Diameter of single contact point C (mm)	d_C	76.0577	103.9455
Diameter of single contact point D (mm)	d_D	77.7294	102.4443
Diameter of single contact point E (mm)	d_E	81.3660	100.0357
Transverse contact ratio	ϵ_{α}	1.5337	
Overlap ratio	ϵ_{β}	0.0000	
Total contact ratio	ϵ_{γ}	1.5337	
Total contact ratio with tolerance	e/m/i	1.476 / 1.454 / 1.433	
Specific sliding at Tip	ζ_a	0.5138	0.5039
Specific sliding at Root	ζ_f	-1.0158	-1.0570

2.5. Calculated geometry - Accuracy

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Quality Standard	-	ISO 1328: 2013	ISO 1328: 2013
Single pitch deviation (μm)	f_pt	8.5000	8.5000
Base circle pitch deviation (μm)	f_pb	8.0000	8.0000
Sector pitch deviation over k/8 pitches (μm)	F_pk/8T	18.0000	18.0000
Profile form deviation (μm)	f_fa	9.0000	9.0000
Profile slope deviation (μm)	f_Ha	7.0000	7.0000
Total profile deviation (μm)	F_α	11.0000	11.0000
Helix form deviation (μm)	f_fb	11.0000	11.0000
Helix slope deviation (μm)	f_Hβ	9.5000	9.5000
Total helix deviation (μm)	F_β	15.0000	15.0000
Cumulative pith deviation, total (μm)	F_p	26.0000	28.0000
Runout (μm)	F_r	23.0000	25.0000
Single flank composite, tooth-to-tooth (μm)	f_isT	8.5000	8.5000
Single flank composite, total (μm)	F_isT	34.5000	36.5000
Tooth-to-tooth radial composite deviation (μm)	f_idT	9.5000	9.5000
Total radial composite deviation (μm)	F_idT	31.0000	31.0000

2.6. Calculated geometry - Measurement

Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Number of teeth spanned	k	4.0000	5.0000
Base tangent length (mm)	W_k	27.4172	35.0111
Base tangent length with tolerance (mm)	W_k, e/i	27.3514 / 27.3138	34.9453 / 34.9077
Tolerance of W_k (mm)	ΔW_k , e/i	-0.0658 / -0.1034	-0.0658 / -0.1034
Diameter of contact point (mm)	d_M	75.5915	102.4554
Theoretical diameter of ball/pin (mm)	D_M.theory	4.5074	4.3504
Effective Diameter of ball/pin (mm)	D_M.eff	5.0000	4.5000
Radial single ball measurement (mm)	M_rk	42.2644	55.0543
Radial single ball measurement with tolerance (mm)	M_rk, e/i	42.1932 / 42.1523	54.9738 / 54.9276
Diametral two ball measurement (mm)	M_dk	84.5288	110.0312
Diametral two ball measurement with tolerance (mm)	M_dk, e/i	84.3863 / 84.3045	109.8703 / 109.7778
Tolerance of M_dk (mm)	ΔM_{dk}	-0.1424 / -0.2243	-0.1609 / -0.2534

3. Detail rating

3.1. General

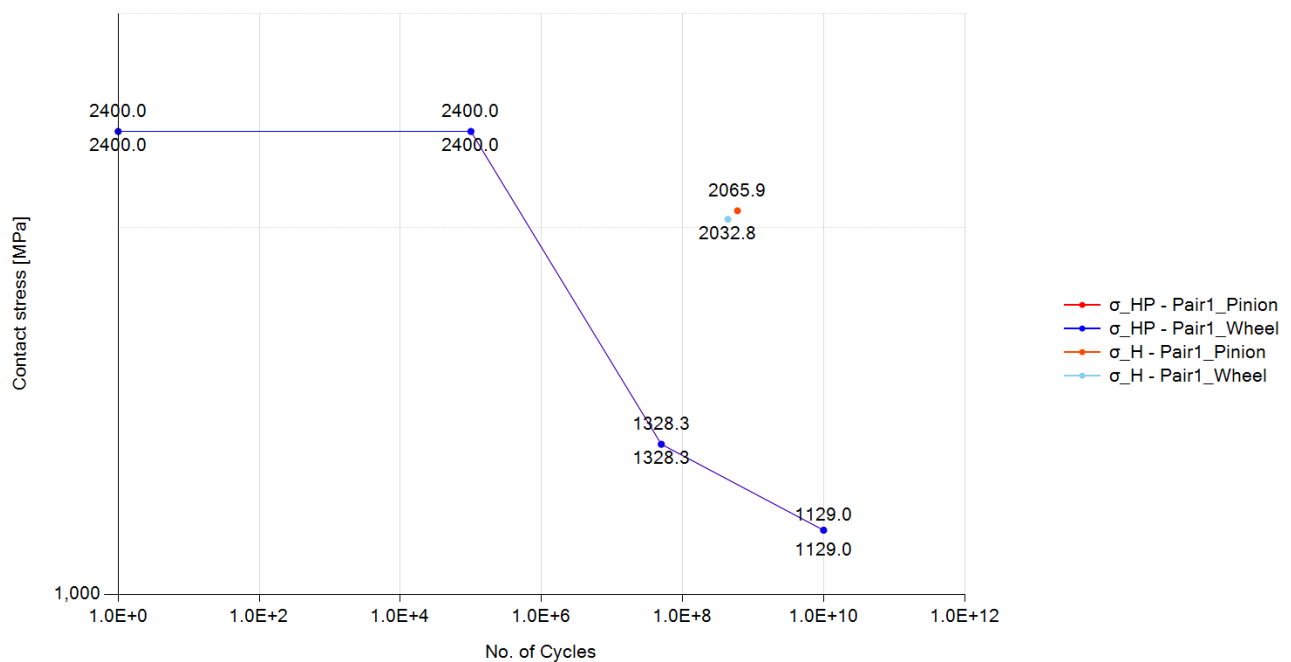
Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rotational speed (rpm)	ω	1000.0000	731.7070
Nominal tangential load at d (N)	F_t	29268.2930	
Nominal axial load at d (N)	F_a	0.0000	
Nominal radial load at d (N)	F_r	10652.7870	
Nominal normal load at d (N)	F_n	31146.6670	
Nominal tangential load per unit facewidth (N/mm)	w	650.4070	
Circumferential speed at d (m/s)	v	3.9270	
Sliding velocity at Tip dia. (m/s)	v_g	1.0940	0.9590
Nominal tangential load at d_w (N)	F_{tw}	28861.2800	
Nominal axial load at d_w (N)	F_{aw}	0.0000	
Nominal radial load at d_w (N)	F_{rw}	11710.7370	
Circumferential speed at d_w (m/s)	v_w	3.9820	
Running-in value (μm)	y_p	0.6000	
Running-in value (μm)	y_f	0.6750	
Effective single pitch & profile deviation (μm)	f_{pb_eff}	7.4000	
Effective single pitch & profile deviation (μm)	f_{fa_eff}	8.3250	
Correction factor	C_M	0.8000	
Gear blank factor	C_R	0.8980	
Basic rack factor	C_B	0.9750	
Material factor	E/E_{st}	1.0000	
Theoretical single stiffness (N/mm/ μm)	c'_{th}	18.3710	
Single tooth stiffness (N/mm/ μm)	c'	12.8680	
Mesh stiffness (for K_v , $KH\alpha$, $KF\alpha$), (N/mm/ μm)	$c_{y\alpha}$	18.0180	
Mesh stiffness (for $KH\beta$, $KF\beta$), (N/mm/ μm)	$c_{y\beta}$	15.3160	
Mean diameter for K_v (mm)	$d_m (K_V)$	75.9410	102.9390
Moment of inertia for K_v ($\text{kg}\cdot\text{mm}^2$)	$J (K_V)$	1004.6060	2938.2210
Moment of gear inertia per unit facewidth ($\text{kg}\cdot\text{mm}$)	J^*	22.3250	65.2940
Relative gear mass per unit facewidth (kg/mm)	m^*	0.0180	0.0282
Relative gear mass per unit facewidth (kg/mm)	m_{red}	0.0110	
Resonance running speed (rpm)	n_{E1}	12899.4170	
Resonance ratio	N	0.0780	
Resonance ratio in main resonance ratio	N_s	0.8500	
Resonance range	-	subcritical range	
Dynamic factor	K_V	1.0240	

	Symbol	Pair1 Gear1	Pair1 Gear2
Face load factor - flank	$K_{H\beta}$	1.3000	
Face load factor - root	$K_{F\beta}$	1.2580	
Transverse load factor - flank	$K_{H\alpha}$	1.0000	
Transverse load factor - root	$K_{F\alpha}$	1.0000	
Number of load cycles	N_L	6.0000×10^8	4.3900×10^8

3.2. Pitting

S-N curve (Pitting), Load case 1, Meshing pair: Gear1 - Gear2



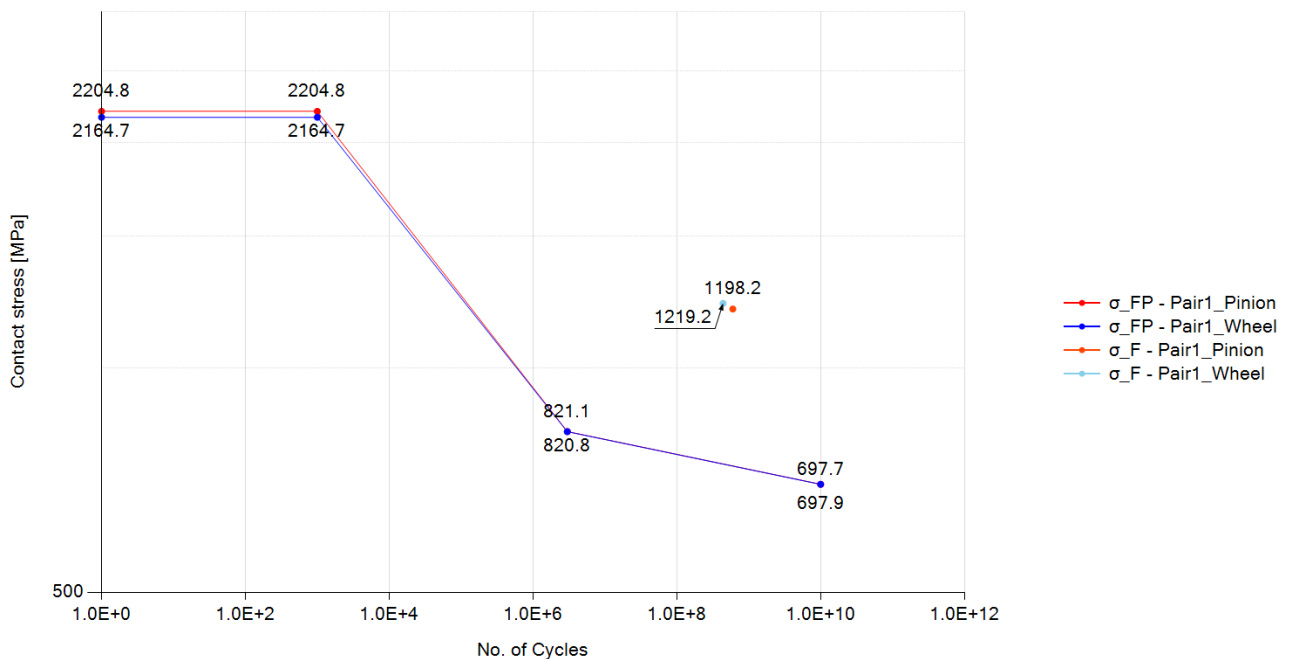
Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rating Method	-	ISO 6336-2:2019	
Contact ratio factor	Z_{ϵ}	0.9070	
Zone factor	Z_H	2.3630	
Elasticity factor	Z_E	189.8120	
Young's modulus for tooth flank (N/mm^2)	E	2.0600×10^5	2.0600×10^5
Helix angle factor	Z_{β}	1.0000	
Single pair tooth contact factor (pinion)	Z_B	1.0160	
Single pair tooth contact factor (wheel)	Z_D	1.0000	
Nominal contact stress at the pitch point (N/mm^2)	σ_{H0}	1575.7290	
Contact stress (N/mm^2)	σ_H	2065.9330	2032.8370
Life factor	Z_{NT}	0.9270	0.9360

	Symbol	Pair1 Gear1	Pair1 Gear2
Lubricant factor	Z_L	0.9460	0.9460
Velocity factor	Z_V	0.9770	0.9770
Roughness factor	Z_R	0.9580	0.9580
Work hardening factor	Z_W	1.0000	1.0000
Size factor	Z_X	1.0000	1.0000
Permissible contact stress - static (N/mm ²)	σ_{HP_static}	2400.0000	2400.0000
Permissible contact stress - mid point (N/mm ²)	σ_{HP_mid}	0.0000	0.0000
Permissible contact stress - reference (N/mm ²)	σ_{HP_ref}	1328.2710	1328.2710
Permissible contact stress - long (N/mm ²)	σ_{HP_long}	1129.0300	1129.0300
Permissible contact stress (N/mm ²)	σ_{HP}	1230.7900	1242.6400
Safety factor for Contact	S_H	0.5960	0.6110
Contact damage (%)	-	1.2426×10^5	76735.0010
Cycle to failure	-	4.8290×10^5	5.7210×10^5

3.3. Bending

S-N curve (Bending), Load case 1, Meshing pair: Gear1 - Gear2



Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Rating Method	-	ISO 6336-3:2019	
Tooth form factor	Y_F	1.3900	1.4440
Stress correction factor	Y_S	2.0580	2.0160

	Symbol	Pair1 Gear1	Pair1 Gear2
Working angle (deg)	α_{Fen}	22.4480	21.6470
Bending moment arm (mm)	h_{Fe}	2.7680	2.9020
Tooth thickness at root (mm)	s_{Fn}	5.4210	5.4620
Tooth root radius (mm)	ρ_F	1.1850	1.2150
Helix angle factor	Y_β	1.0000	
Deep tooth factor	Y_{DT}	1.0000	
Rim thickness factor	Y_B	1.0000	1.0000
Nominal tooth root stress (N/mm ²)	σ_{F0}	744.0080	757.0160
Tooth root stress (N/mm ²)	σ_F	1198.2070	1219.1560
Life factor	Y_{NT}	0.8990	0.9050
Relative notch sensitive factor	$Y_{\delta relT}$	0.9980	0.9980
Surface factor	Y_{RrelT}	0.9570	0.9570
Size factor	Y_X	1.0000	1.0000
Stress correction factor	Y_{ST}	2.0000	
Mean stress influence factor	Y_M	1.0000	1.0000
Technical factor	Y_T	1.0000	1.0000
Permissible tooth root stress - static (N/mm ²)	σ_{FP_static}	2204.7700	2164.7330
Permissible tooth root stress - reference (N/mm ²)	σ_{FP_ref}	821.1120	820.7970
Permissible tooth root stress - long (N/mm ²)	σ_{FP_long}	697.9450	697.6770
Permissible tooth root stress (N/mm ²)	σ_{FP}	738.4160	742.7670
Safety factor for Bending	S_F	0.6160	0.6090
Bending damage (%)	-	6.5457×10^6	6.1708×10^6
Cycle to failure	-	9166.0000	7115.0000

3.4. Scuffing

Load case 1, Meshing pair: Gear1 - Gear2

	Symbol	Pair1 Gear1	Pair1 Gear2
Lubrication type	-	Spray lubrication	
Lubricant temperature (deg)	θ_{oil}	80.0000	
Lubricant viscosity (mPa.s)	η_{oil}	13.1510	
Load stage according to FZG A/8, 3/90	S_{FZG}	12.0000	
Gear density (kg/m ³)	ρ_M	7830.0000	7830.0000
Gear specific heat (J/kg.K)	c_M	485.0000	485.0000
Gear heat conductivity (N/s.K)	λ_M	50.0000	50.0000
Gear roughness (μm)	R_a	0.6000	0.6000
Number of mating gears	n_p	1.0000	
Effective facewidth (mm)	b_{eff}	38.2500	

	Symbol	Pair1 Gear1	Pair1 Gear2
Relevant tip relief (μm) (Min. value = 2 μm)	C_a	2.0000	2.0000
Transverse unit load (N/mm)	w_Bt	1082.4970	
Lubrication system factor	X_S	1.2000	
Multiple meshing factor	X_mp	1.0000	
Structural factor	X_W	1.0000	1.0000
Structural factor (total)	X_Wtot	1.0000	
Thermal contact factor	B_M	13.7800	13.7800
Multiple path factor	K_mp	1.0000	
Optimal tip relief (μm)	C_eff	63.1810	
Angle factor	X_αβ	1.0090	
Lubricant factor	X_L	0.8790	
Roughness factor	X_R	0.7750	
Mean coefficient of friction	μ_m	0.0870	
Maximum flash temperature (deg)	θ_fl_max	67.7470	
Mean flash temperature (deg)	θ_fl_mean	42.3700	
Total mass temperature (deg)	θ_M	103.8970	
Scuffing temperature (deg)	θ_S	364.8370	
Maximum contact temperature (deg)	θ_Bmax	171.6440	
Safety factor for flash temperature	S_B	3.1080	

3.5. Micro-pitting

Load case 1

	Symbol	Gear1 - Gear2
minimum lubricant film thickness (μm)	h_Ymin	0.0393
minimum specific lubricant film thickness in the contact area	λ_GFmin	0.0654
permissible specific lubricant film thickness	λ_GFP	0.2111
safety factor against micropitting	S_λ	0.3099
transverse tangential load at reference cylinder per mesh (N)	F_t	29268.2930
transverse load in plane of action (N)	F_bt	31146.6670
reduced modulus of elasticity (N/mm ²)	E_r	2.2637×10 ⁵
thermal contact coefficient of pinion (N/(ms ^{0.5} *K))	B_M1	13779.6040
thermal contact coefficient of wheel (N/(ms ^{0.5} *K))	B_M2	13779.6040
failure load stage according to FVA 54/7 (90°)	-	0.0000
effective arithmetic mean roughness value (μm)	R_a	0.6000
roughness factor	X_R	1.1420
lubricant factor	X_L	1.0000
lubrication factor	X_S	1.2000

	Symbol	Gear1 - Gear2
elasticity factor (N/mm ²) ^{0.5}	Z_E	189.8120
helical load factor	K_Bγ	1.0000
load losses factor	H_v	0.1170
mean coefficient of friction	μ_m	0.0980
profile position at min. lubricant film thickness	-	0.0000
tip relief factor	X_Ca	1.0000
load sharing factor	X_A	0.3330
buttressing factor	X_but,A	0.0000
effective tip relief (μm)	C_eff	0.0000
Applied tip relief in calculation (μm)	C_a	0.0000
sum of tangential velocities at point A (m/s)	v_Σ,A	2.8460
sliding velocity at point A (m/s)	v_g,A	-0.9590
tangential velocity on pinion at point A (m/s)	v_r1,A	0.9440
tangential velocity on wheel at point A (m/s)	v_r2,A	1.9020
oil inlet/sump temperature (°C)	θ_oil	80.0000
bulk temperature (°C)	θ_M	102.8180
flash temperature at point A (°C)	θ_fl,A	68.1310
contact temperature at point A (°C)	θ_B,A	170.9500
pressure-viscosity coefficient at 38 °C (m ² /N)	α_38	1.8390×10 ⁻⁸
pressure-viscosity coefficient at bulk temperature (m ² /N)	α_θM	1.3130×10 ⁻⁸
pressure-viscosity coefficient at point A contact temperature (m ² /N)	α_θB,A	9.2530×10 ⁻⁹
kinematic viscosity at 40 °C (mm ² /s)	ν_40	68.0000
kinematic viscosity at 100 °C (mm ² /s)	ν_100	8.8000
kinematic viscosity at point A contact temperature (mm ² /s)	ν_θB,A	2.6320
dynamic viscosity at 38 °C (Ns/m ²)	η_38	0.0650
dynamic viscosity at bulk temperature (Ns/m ²)	η_θM	0.0070
dynamic viscosity at oil inlet/sump temperature (Ns/m ²)	η_θoil	0.0120
dynamic viscosity at point A contact temperature (Ns/m ²)	η_θB,A	0.0020
normal radius of relative curvature at point A (mm)	ρ_n,A	6.6120
material parameter	G_M	2971.9670
local velocity parameter at point A	U_A	6.4560×10 ⁻¹²
local load parameter at point A	W_A	0.0003
local sliding parameter at point A	S_GF,A	0.2120
local nominal Hertzian contact stress at point A (Method B) (N/mm ²)	p_H,A	1121.2290
local Hertzian contact stress at point A including the load factors K (Method B) (N/mm ²)	p_dyn,A	1446.4900

4. Power loss

4.1. Mesh loss

Load case 1

	Symbol	Gear1 - Gear2
Rating Method	-	ISO/TR 14179-2:2001
Input Power (kW)	P	114.9363
Gear ratio	u	1.3667
Tangential speed at working pitch dia. (m/s)	v_{wt}	3.9824
Tangential force at pitch dia. (N)	F_t	29268.2927
Tooth normal force, transverse section (N)	F_{bt}	31146.6665
Arithmetic average roughness of pinion	Ra1	0.6000
Arithmetic average roughness of wheel	Ra2	0.6000
Roughness factor	X_R	0.6000
Sum velocity (m/s)	v_{Σ}	2.9946
Equivalent radius of curvature (mm)	ρ	8.2568
Kinematic viscosity of oil at operating temp. (cSt)	η_{oil}	14.8602
Lubricant type	-	Mineral oil base
Lubricant factor	X_L	1.0000
distributed force on tooth surface(N/m)	F/b	692.1481
Mean coefficient of friction of the gear mesh	μ_{mz}	0.0719
Tooth loss factor	H_v	0.1174
Mesh power loss (W)	P_{VZP}	970.0643
Mesh efficiency (%)	η_z	99.1560

4.2. Churning loss

Load case 1

	Symbol	Gear1	Gear2
Rating Method	-	Changenet (2011)	Changenet (2011)
Rotating speed (rpm)	n	1000.0000	731.7073
Lubricant	-	Oil: ISO-VG 68	Oil: ISO-VG 68
Gear immersed depth (mm)	h_e	38.2830	51.7822
Oil density (kg/m ³)	ρ	885.0000	885.0000
Operating Viscosity (cSt)	ν	14.8602	14.8602
dynamic viscosity (kg/m.s)	μ	0.0000	0.0000
Reynolds number	Re	12140.1081	12015.2848
Froude number	Fr	42.7951	30.9915
Centrifugal acceleration (m/s ²)	γ	178.4199	105.6435

	Symbol	Gear1	Gear2
Tip diameter of gear (mm)	d_a	81.5660	108.5643
Pitch diameter of gear (mm)	d_p	76.5660	103.5643
Facewidth (mm)	b	45.0000	45.0000
Dimensionless drag torque	C_m	0.0242	0.0252
Immersion area of the gear (m^2)	S_m	0.0181	0.0268
Total oil churning loss (W)	P	12.4785	18.6303
Total oil churning torque (N.m)	T	0.1192	0.2431

4.3. Windage loss

Load case 1

	Symbol	Gear1	Gear2
Rating Method	-	None	None
Windage loss (W)	P_WL	0.0000	0.0000
Windage torque (N.m)	T_WL	0.0000	0.0000